

# Nuclear Bomb And RADIATION DETECTION DEVICE

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Nuclear radiation is one of the most dangerous killers for the living. The atomic radiation needs to be detected at an early stage, otherwise it can be devastating within its area of influence. The radiation kills not only

humans, it also destroys all natural habitat. It is important to avoid harm and mass destruction that can be caused by nuclear weapons and reactors nearby.

So, let us design a nucle-

## BILL OF MATERIAL

| Components            | Quantity | Description           |
|-----------------------|----------|-----------------------|
| GM Muller tube        | 1        | M4011 GM tube         |
| Geiger counter module | 1        | 5V Geiger counter     |
| SSD1306 OLED          | 1        | OLED display          |
| ESP32 S board         | 1        | Microcontroller board |

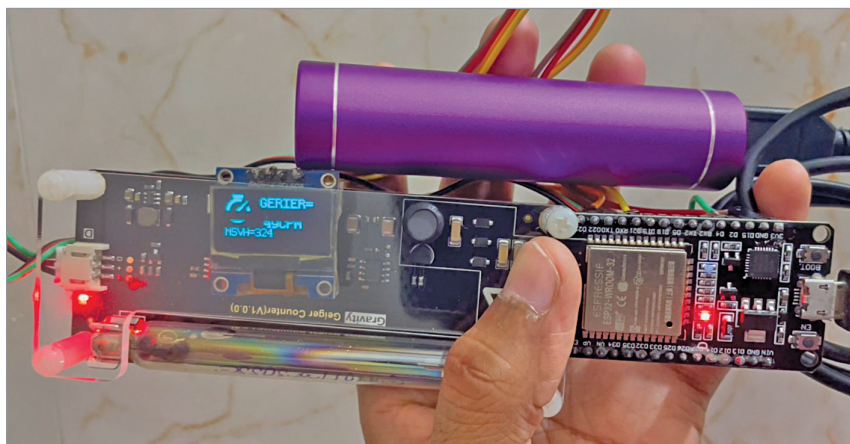


Fig. 1: Author's prototype

ar radiation detector that can measure and detect the alpha, beta, and gamma radiations and alert us well in time if the radiation is too high and harmful for us. Here we use the Muller tube and a Geiger counter circuit to detect and measure the nuclear radiation. The author's prototype is shown in Fig. 1.

Fig. 2 shows the circuit diagram of nuclear bomb and radiation detection device. The circuit is built around GM Muller tube, Geiger counter module, SSD1306 OLED, and ESP32S board.

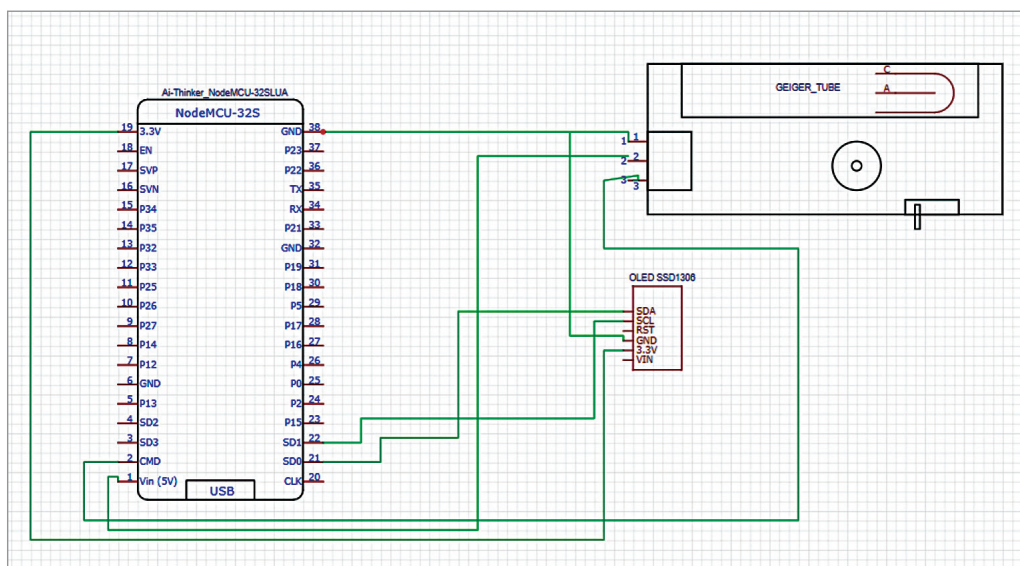


Fig. 2: Nuclear bomb and radiation detection device circuit

## Software

First install the ESP32 board in Arduino IDE using the instructions provided at link <https://espressif-docs.readthedocs-hosted.com/projects/arduino-esp32/en/latest/installing.html>.

Next, install the library for the Geiger module being used. (The author used the DF robot Geiger, so he installed the library for that.) In the code set the pin for reading the sensor data from